Game of Thrones Book Q&A; Chatbot

Final Architecture — Embedding + Traditional Retrieval + SLM/LLM

1. Objective

Build a question-answering chatbot using the **Game of Thrones (GOT) book content as the only knowledge source**. Users can ask about characters, events, relationships, plots, actions, and other information contained in the book. The system must refuse or return an unknown response when the requested information is outside the book knowledge base.

2. Final Architecture

GOT Book Authoritative knowledge source
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1. Text Extraction & Cleaning Extract book text • normalize text • preserve chapter/source metadata
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2. Chunking & Metadata Create meaningful passages while retaining book, chapter and passage references
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3. Embedding Model Convert passages into semantic vector representations
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4. Vector Index / Database Store embeddings and source metadata for retrieval
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User Question
↓
5. Query Embedding Convert the question into the same semantic vector space
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6. Traditional / Semantic Retrieval Retrieve Top-K relevant GOT passages
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7. Domain & Evidence Gate Determine whether relevant GOT evidence exists and sufficiently supports the question
■

Outside Knowledge / Insufficient Evidence Refuse or return: “No known answer in the GOT book.”
■
8. SLM / LLM Generate an answer using ONLY the retrieved GOT passages
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9. Final Answer Answer + optional chapter/passage reference

3. Knowledge Boundary

The GOT book is the **only source of truth**. The pretrained SLM/LLM is not permitted to supply factual knowledge from its general training. Its role is limited to understanding the question and generating a natural-language answer from retrieved GOT evidence.

4. Expected Behaviour

Question	Expected result
Who killed [a character]?	Answer if the event is supported by retrieved GOT passages.
Who married [a character]?	Answer from the book evidence if the relationship is documented.
What happened during [an event]?	Summarize the relevant retrieved passages.
Who is Harry Potter?	Refuse / “No known answer in the GOT book.”
What happened in a book not included?	Refuse / outside the knowledge boundary.

5. Strict Answering Flow

Question → **Retrieve** → **Validate evidence** → **Generate answer**. The evidence gate is essential: do not rely only on an LLM instruction such as “answer only from the book.” If retrieval does not provide sufficiently strong evidence, the generation step is skipped.

6. Example

User: Who married Robb Stark?
Retrieval: Relevant GOT passages are retrieved.
Evidence Gate: Checks whether the passages actually support the relationship.
SLM/LLM: Generates the answer only from those passages.
Final: Answer with an optional chapter/passage reference.

7. Recommended Technology Pattern

Use a pretrained embedding model for semantic retrieval, a vector index/database for efficient Top-K search, and a small language model (SLM) or larger LLM purely as the answer-generation layer. This keeps the architecture modular: each component can be replaced independently.

8. Core Design Principle

Retrieve first → validate evidence → generate second.
This architecture provides a controlled, book-grounded Q&A; system while minimizing hallucination and preventing answers based on knowledge outside the GOT book.